LEVELHULL · DESIGN DECISIONS REGISTER

LevelHull design decisions register

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DOCUMENT CONTROL

Document ID	LVH-DEC-001	Version	0.4
Status	Draft	Date	2026-10-25
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/levelhull		
Source	docs/06-design-decisions.md @ e0fd582		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-25	Amish Chadha	Amish decided D-O2 (b), D-O3 (a), D-R1 and the product model appearance departures as recommended on 2026-10-25 ("i concur with your recommendations"); D-O1 stays open (no pick); USD 1,384
0.3	2026-10-24	Amish Chadha	Kit 1.9.1 upgrade: D-O1 concept choice (A or B), D-O2 sealed hull after swamping, D-O3 bailing and D-R1 new requirements PR-014 to PR-017 open, awaiting Amish; D-S1 end chock screws and D-S2 coach screw material decided under the standing instruction of 2026-10-23; calculation corrections from the independent review; kit 1.9 detailing recorded with no design dimension changed
0.2	2026-10-03	Amish Chadha	Decision 10A recorded (modules widened for R12 and R13, LVH-DDR-003); value engineering updated
0.1	2026-10-03	Amish Chadha	Register opened; all decisions made under Amish's 2026-10-03 pre-approval

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Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

> **Safety:** Several entries below set safety limits (where the modules sit, no credit for the wood, screws kept out of the planking, the pull test and the swamp test rules). Each takes the conservative option and names the evidence that would relax it. LevelHull is never a substitute for life jackets.

Open decisions

Table 0. Open decisions, Proposed, awaiting Amish. D-O2, D-O3 and D-R1 were decided by Amish on 2026-10-25 as recommended and are in Table 2.

#	To decide	Options	Recommendation	What it affects in the build	Source
D-O1	Concept choice	A: foam modules high inside (the design, 172 of 215); B: external foam sponsons (163); C: double outrigger floats (148, not put forward). Variant of A: lower the modules and battens about 40 mm (B's lesson)	None: A and B are within 10 % (5.2 %), so both go to Amish without a pick. The repo stays on A until he decides	If B: new model detail for construction, drawings, BOM and build plan; if the A variant: batten height, gauge notches, sizing re-run	LVH-OPT-001, LVH-DDR-004

To confirm when parts are bought

These are facts that can only be settled with real parts, a real canoe or the first partner. None changes a decision; each may change a size or a limit.

Table 1. Items to confirm.

#	What to confirm	Why it matters	Source
1	Foam: density about 30 kg/m3, water uptake under 1 % by volume, fuel resistance, on the supplier's data sheet; then 30 days immersion of samples	R6, and every freeboard figure	LVH-CAL-001, A6; LVH-DDR-002, A4
2	First canoes: frame spacing (600 mm assumed) and depth (60 mm), side slope, thwart positions, timber, gunwale and strake sizes	Batten height, screw length, module lengths and the gauge	LVH-DDR-002, A1
3	Hull mass and timber volume of the first canoes	Size of the unclaimed wood margin (D3)	LVH-DDR-001, D3
4	Condition of the frame-to-plank fastenings	Each frame takes up to 0.19 kN of lift	LVH-DDR-002, A3
5	Webbing breaking strength (10 kN assumed), D-ring pair (4 kN) and coach screw sideways strength in the canoe's timber (1.5 kN)	R7 factors	LVH-CAL-001, section 7
6	Real motor mass and the gear, catch and crew of the partner's fishery	Design net load of 118 kg	LVH-CAL-001, A3 to A5
7	Bailing rate with the local bailer (100 L per minute per person assumed)	R4	LVH-CAL-001, A8
8	Price of polyethylene foam sheet near the first fishery (USD 55 a sheet assumed)	Foam is 48 % of the kit cost	`bom/bom.csv`, line 5

Value engineering

Value-engineering target: USD 1,000 (a hypothetical control target, not a limit) for the kits for two prototype canoes. Estimated cost of the constructable design: USD 1,384 (USD 384 over the target); one kit is USD 694, with four bailers since D-O3 (USD 12 more a kit). Main cost drivers and savings worth trying:

- The largest lines are the foam (USD 330 a kit, six sheets), the tarpaulin covers and their sewing parts (USD 100), the battens (USD 56) and the coach screws (USD 43).
- Savings worth trying: buy foam by the block from a fishing-float or packaging maker and cut it locally (perhaps USD 100 a kit); galvanised D-rings in place of stainless where the canoe is used in fresh water (the screws stay A4, D-S2); covers from used truck tarpaulin in good condition (about USD 50); a three-layer module on hulls whose weighed timber gives a proven wood margin (D3), cutting foam by a quarter.
- The gauge is made once per hull shape and the calculator is free, so a builder fitting a fleet of the same canoe spreads those costs.

Decisions made

Table 2. Decisions made.

Date	Decision	Decided by	Record
2026-10-25	D-O2, the sealed hull just after swamping: option (b), keep the hull sealed and state R2 and R3 for the free-flooding condition of the USCG method, with the operating rule "hold on, then bail at once, seated"; one-way freeing ports (a) studied at TRL 4 with the partner	Amish: "i concur with your recommendations"	LVH-PRS-001 v0.5 PR-002, PR-003; LVH-STM-001 v0.2 S32; LVH-BLD-001 v0.4 step 10
2026-10-25	D-O3, bailing from brim-full: option (a), four bailers so all four crew bail (BOM line 12, 2 to 4, USD 12 more a kit); about 11 min from brim-full at 50 L/min each against 15	Amish: "i concur with your recommendations"	BOM line 12; `cad/src/model.py` `bailers_x`; LVH-PRS-001 v0.5 PR-004; LVH-STM-001 v0.2 S34; LVH-CAL-001 v0.4
2026-10-25	D-R1, requirements added in the kit 1.8 migration: PR-014 to PR-017 accepted as written	Amish: "i concur with your recommendations"	LVH-PRS-001 v0.5
2026-10-25	Appearance departures of the kit 1.9 product model (renders only): beach slab, 1.75 m mannequin, outboard tilted up, canoe topsides painted blue, as proposed on 2026-10-24	Amish: "i concur with your recommendations"	`cad/src/product_model.py` (no change)
2026-10-24	D-S1, end chock screws: the two 5 x 60 mm stainless screws in each chock went straight down through the 50 mm chock and only 10 mm into the batten (LVH-STM-001 S13 failed: 6 d is 30 mm). Options weighed: longer 5 x 80 mm screws straight down (would break out of the batten's sloping inboard face below about 30 mm, and cost more); one M8 coach screw per chock (chock too small); screws driven parallel to the frame face in 12 mm counterbores 30 mm deep (chosen: same screws, same chock, the screw stays inside chock and batten for its whole length, 31 mm into the batten). Cost change: none. Model, BOM line 4, build plan (section 3.3, step 4) and LVH-DWG-103 updated; S13 rerun and closed	Decided under Amish's standing instruction of 2026-10-23 (fix a failed STEM row in the same session); Amish can reverse it	LVH-STM-001 S13
2026-10-24	D-S2, coach screw material: the BOM allowed hot-dip galvanised coach screws as an alternative to A4 stainless; tannins in wet hardwood attack zinc in fresh water as well as salt (independent review of LVH-STM-001 S22). Galvanised withdrawn; A4 only, inspected for crevice corrosion at each refit. Cost change: none (the BOM was priced for stainless)	Decided under Amish's standing instruction of 2026-10-23; Amish can reverse it	LVH-STM-001 S22; BOM line 3
2026-10-24	Calculation corrections from the independent review (no design change): the floor comparison of LVH-CAL-001 section 5 used the high modules' waterplane position (now 135 kg m per radian, 22 deg with the crew on one side, was 283 and	Corrections, not decisions; recorded for the audit trail	LVH-CAL-001 v0.3

Date	Decision	Decided by	Record
	10.6 deg); denser-than-water timber now carries its weight moment in the trim; fitting time 6.7 h with the counterbored chocks		
2026-10-24	Kit 1.9 detailing of option A, no design dimension changed: holes for every screw, 8.4 x 24 mm washers under the coach screws, every eye bolt with its washers and nyloc nut, foam cores modelled as their four strips, D-ring buckles as wire rings, 2 mm edge breaks on battens and chocks, bailer lanyards, the setting gauge shown beside the canoe	Standing instruction for the kit 1.9.1 rollout (Amish, 2026-10-24: "Detail every option")	`cad/src/model.py`, `cad/fidelity.json`
2026-10-03	TRL 2 review items D1 to D10: modules high under the gunwale, not low; closed-cell polyethylene foam in covers, not chambers or polystyrene; no lift counted from the hull timber; one reference canoe (8.0 m, 15 hp, four crew) with an open calculator for others; crew in the water at 20 % of body weight; re-entry step stays parked; grab lines along both sides; life jacket message in every document and test; first co-design candidates (a beach management unit on the Kenyan shore of Lake Victoria with its canoe builders, a regional fisheries research institute, a sea-safety programme; none approached yet); R11 to R13 added; pitch, problem, design-arounds and budget unchanged	Amish, pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."	LVH-DDR-001
2026-10-03	Design for construction, changes C1 to C12: modules high on the frame faces; four modules a side between thwarts; parallelogram stacks of equal strips; shelf battens; coach screws into frames only, 10 mm short of the planks; end chocks; webbing loops with D-ring buckles in the bays; no glue; sewn tarpaulin covers; setting gauge; grab-line eye bolts above the waterline; freeboard marks and bailers	Amish, same pre-approval	LVH-DDR-002
2026-10-03	Assumptions A1 to A5: frames at most 600 mm apart and 60 mm deep, else shorter screws and a third frame, never through the planks; timber as dense as water; sound frame fastenings checked before fitting; foam data to confirm; fixing strengths to confirm	Amish, same pre-approval	LVH-DDR-002
2026-10-03	No fishing use before the batten pull test, a one-bay proof load to twice the full lift, and a swamp test in sheltered shallow water with a safety boat and every person in a life jacket (conservative; a gate, not relaxed)	Amish, same pre-approval	LVH-BLD-001, sections 5 and 6
2026-10-03		Amish, same pre-approval	LVH-CAL-001, section 4

Date	Decision	Decided by	Record
	Crews spread along both grab lines: with all four on one side the low gunwale comes within 31 mm of the water (44 mm with the wider modules of LVH-DDR-003) (conservative; relaxed only if a swamp test shows more)		
2026-10-03	Appearance model departures: a sand beach slab and a 1.75 m mannequin beside the canoe, and a repeated cut-back module bay for the detail view, drawn for the renders only; canoe topsides shown painted	Amish, same pre-approval	docs/REVIEW.md, TRL 3 section
2026-10-03	Decision 10A for R12 and R13: foam modules enlarged about 10 % by widening them from 200 to 220 mm (height, lengths, positions and fixings unchanged). R12: 25 mm lowest freeboard with the largest module lost (was 9 mm); R13: 3.9 deg heel, 44 mm on the low side (was 4.0 deg, 31 mm); kit 68.9 kg; USD 1,360 for two canoes	Amish, 2026-10-03: "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"	LVH-DDR-003